

Capstone

CS.25.12

Our Team

Team:

- Oscar Eriksen
- Anthony Ly
- John Maciejewski
- Rithikaa Prakash

Advisor:

- Ritika Kanade

Sponsor/ Founder:

- Deborah Boone

Background

company name

- Consignment store
- Founded to pass on tradition
- Sustainability and Skill Preservation

Background

```
// Display Selected Days
if (_selectedDays.isNotEmpty)
  const SizedBox(height: 0),
  SizedBox(
    height: 140, // Adjusted height to fit within screen
    child: SingleChildScrollView(
      padding: EdgeInsets.only(bottom: MediaQuery.of(context).viewInsets.bottom),
      child: Wrap(
        spacing: 8.0, // Space between chips
        children: _selectedDays.map((day) {
          return SizedBox(
            width: 115,
            child: Chip(
              label: Text(DateFormat.MMMd().format(day)), // Formats as "Jan 24"
              deleteIcon: Icon(Icons.close, size: 18),
              onDelete: () {
                setState(() {
                  _selectedDays.remove(day);
                });
              }
            )
          );
        })
      )
    )
  );
```

Technical

- Continuation of last year's project
 - Mobile app based
 - Poor documentation of code
 - Little to no experience with Flutter
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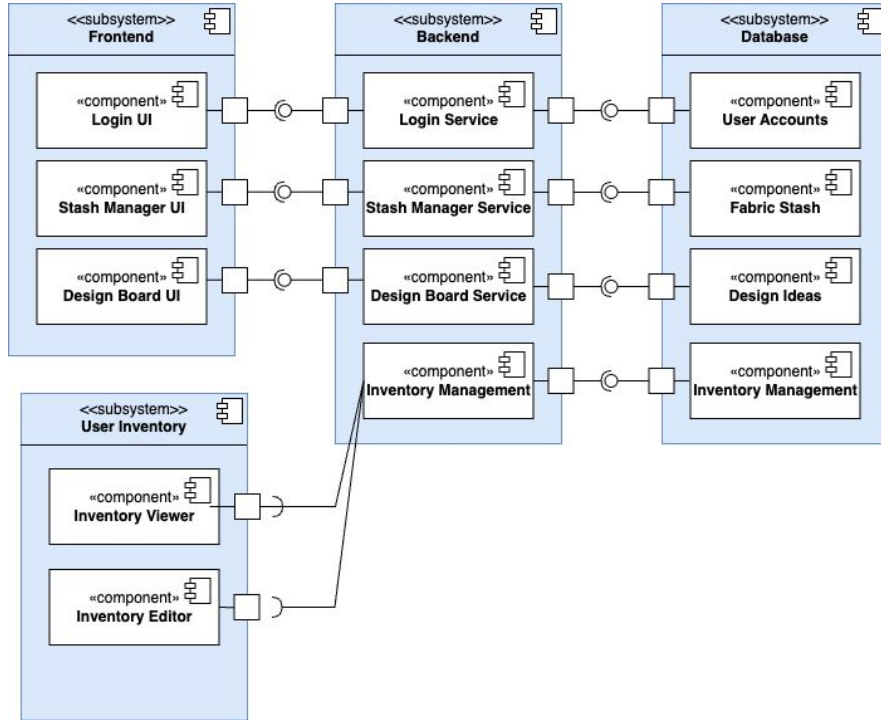
Solution and Key Features

A new app that includes:

- Account creation/management
- Personal Inventory Management
- Class sign-up
- Design boarding



Architecture Diagram



- Frontend UI
 - Login and Authenticator
 - Stash management
 - Design Board tools
- Databases
 - Store account information
 - Store inventory information
 - Store Crafts/Design Ideas
 - Local Cloud Storage
- Backend
 - Authentication details
 - Login service
 - Stash/Design Board/Inventory management
 - Handle API requests
- User Inventory
 - Communicates with backend
 - Handles User Requests

Development Environment



GitHub

Project code management

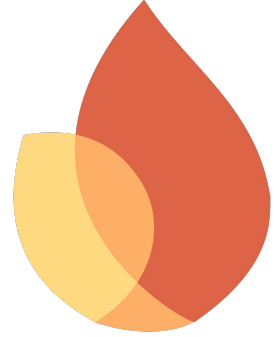


Flutter

Development environment



Dart



FireBase

Backend services

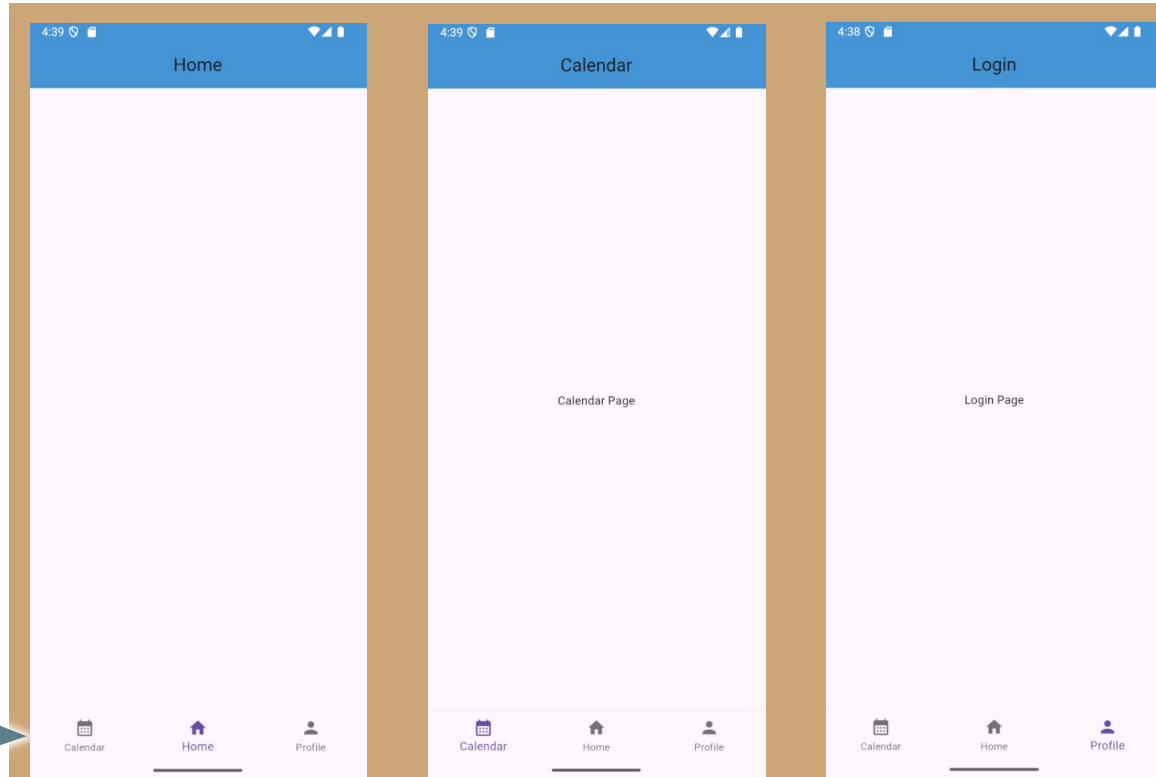
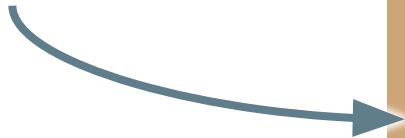


Android Studio

Android virtual phone emulator

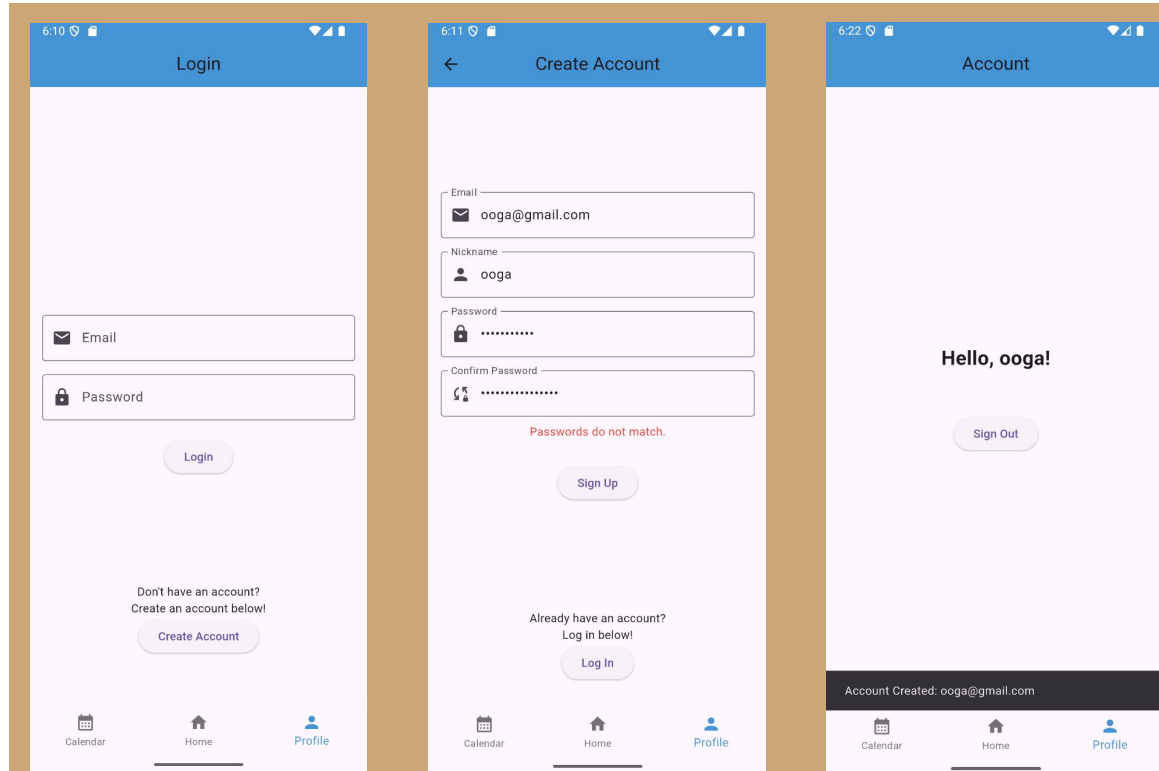
Sprint 1: Set Up App in Flutter and Firebase

- Create Flutter project
- Create Firebase project
- Set up framework
- Add three pages
- Working navigation bar



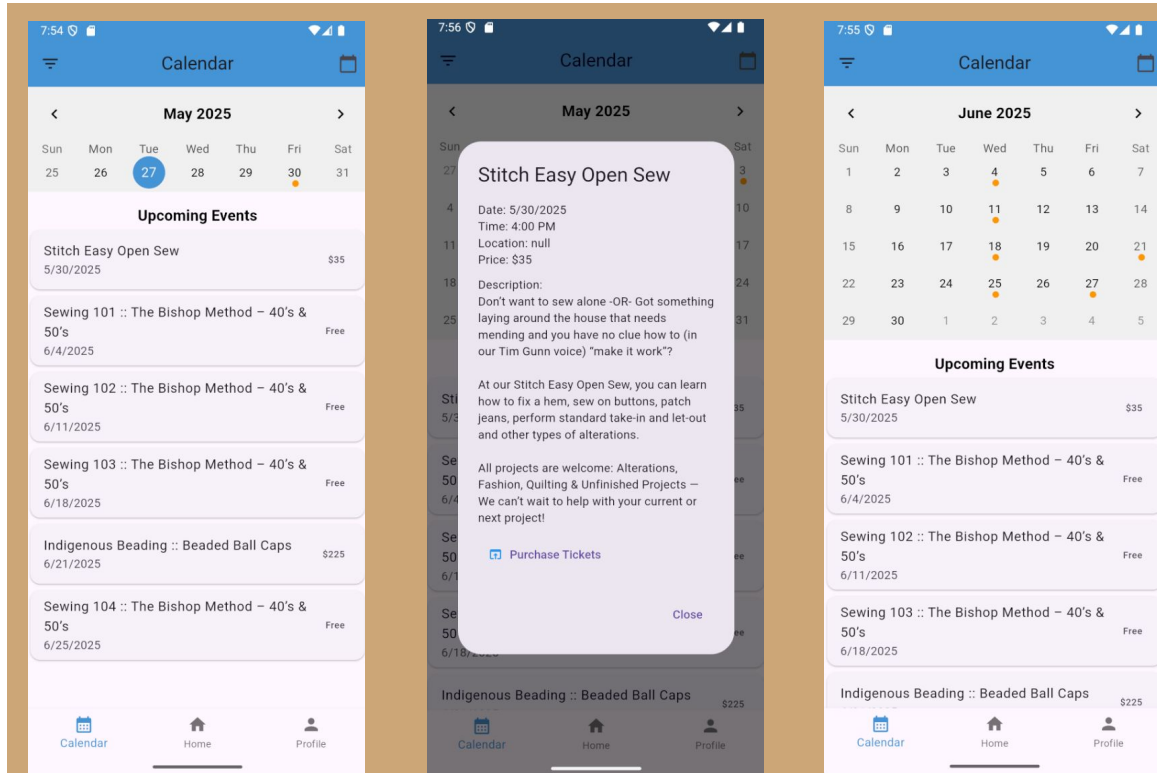
Sprint 2: User Authentication

- Setup database in Firebase
- Connect Firebase to app
- UI Design
- Popup messages



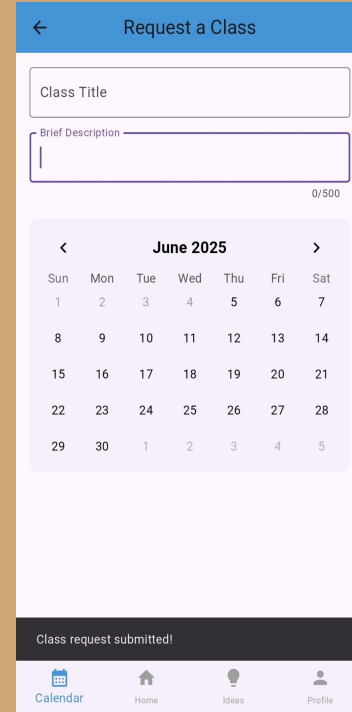
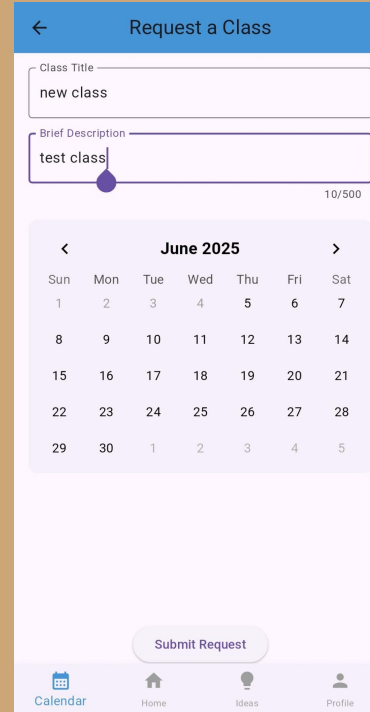
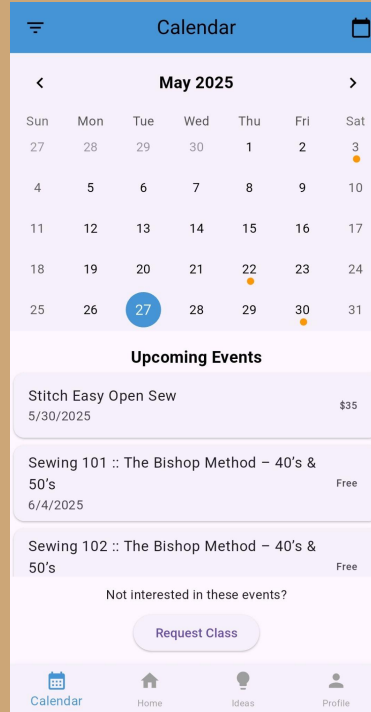
Sprint 3: Interactive Calendar

- Setup and connect calendar to database
- Generate upcoming events from database
- Filter classes by price
- Make calendar and upcoming events clickable
- Connect button to the link to purchase a ticket



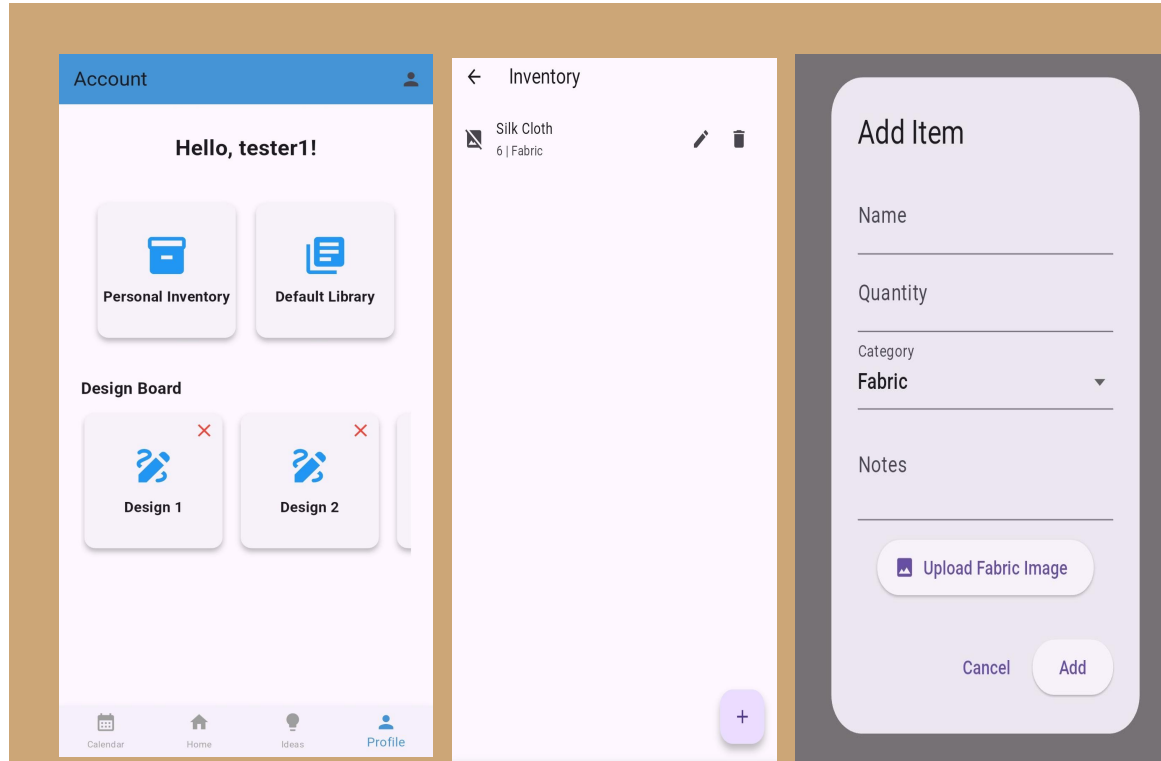
Sprint 4: Class Request Form

- Select Request class in calendar section
- Select date and enter class name and disc.
- Submit Request for new class
- See class request confirmation



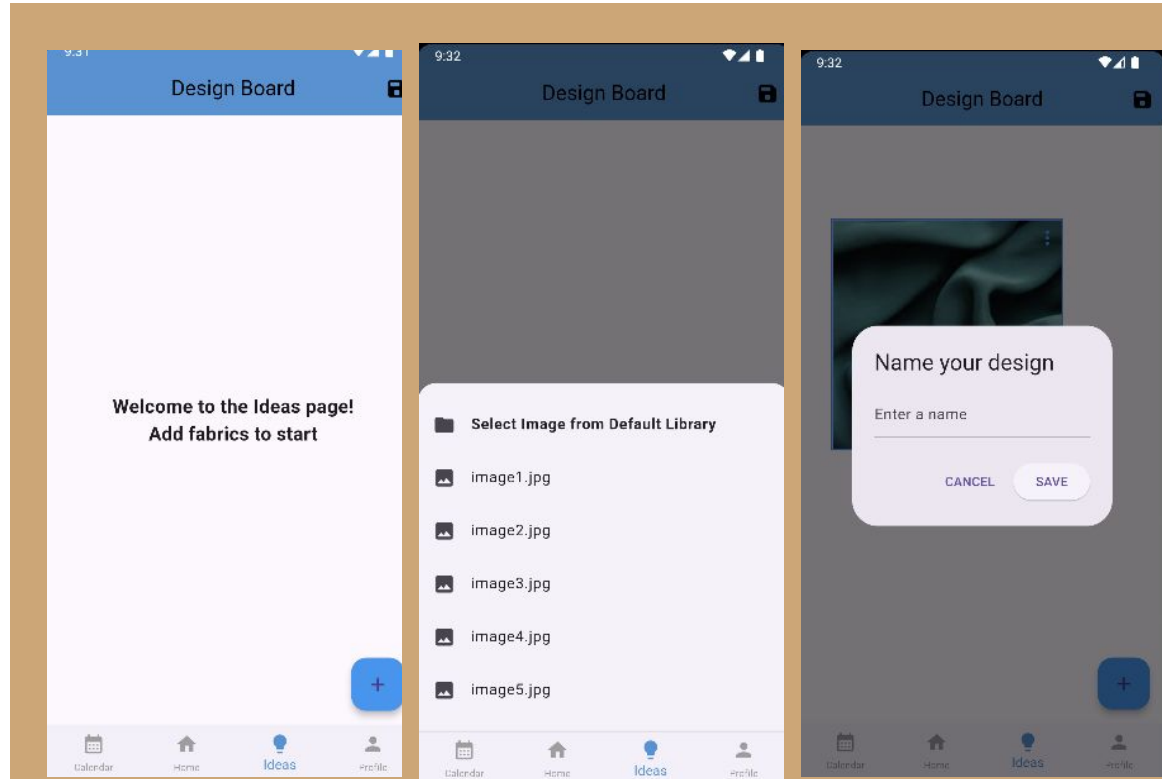
Sprint 5: Personal Inventory System

- Click on Manage Personal Inventory
- To create a new item click the plus icon.
- Fill out info for the item to add to inventory
- Click on pencil icon to edit existing item
- To delete an item click on trash can icon



Sprint 6: Design Board

- Click on the Ideas tab on the navigation bar
- Click the add icon at the bottom left to add an image from the personal inventory
- Drag and resize images
- Click the save icon at the top, name, and save to account screen



Summary of Deliverables



Figure 1

Date	Topic	Description
1/20/25	Set Up App in Flutter and Firebase Sprint	Create our project with the Flutter Framework and attach necessary services to our project so that we won't have to update our app and run into issues later in development. Get account authentication up and running so that we can focus on personalized features in later sprints.
2/13/25	User Authentication Sprint	Utilizing Firebase authentication, create a log in page with all the features that come with that. (mistype check, log out, create account, username, etc.)
3/06/25	Interactive Calendar Sprint	Create a section in the app that shows an interactive calendar displaying the upcoming events and classes that the store is hosting / recommending to customers and fabric lovers.
3/28/25	Class Request Form Sprint	Allows users to request specific classes when the classes that the store is creating don't fit the desires of the customers.
4/18/25	Personal Inventory System Sprint	Tweak last year's personal inventory section in the app to our liking and update the UI to look more user friendly (modern, minimalistic).
5/02/25	Design Board Sprint	Create a design board feature very closely related to the personal inventory feature that will allow users to create physical designs using the materials they have in the personal inventory.
5/09/25	Sponsor Handoff for Final Review	Give Deborah the final version of our app as prepared and ask her for any final tweaks that we should accomplish to make her proud to sign off on our final product.
5/16/25	Sponsor Signoff for Final Solution	Fix any tweaks that Deborah mentioned in the final review and get her signature to sign off on our final product.

Thank you

Bibliography

Github Logo - <https://github.com/logos>

Flutter Logo - <https://icons8.com/icons/set/flutter-logo>

Dart Logo - https://commons.wikimedia.org/wiki/File:Dart_logo.png

Firebase Logo - [https://commons.wikimedia.org/wiki/File:Firebase Logo %28No wordmark%29 %282024-%29.svg](https://commons.wikimedia.org/wiki/File:Firebase_Logo_%28No_wordmark%29_%282024-%29.svg)

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Figure 1 - <https://depositphotos.com/vectors/open-gift-box.html>